

Entity Relationship (ER) Model

↳ Graphical representation of data required for

↳ Building blocks of ER-Model

↳ Entity

↳ Entity type → Set of entities that have same characteristics.

↳ Entity instance → Each occurrence of an entity

↳ Entity set → Collection of instances belonging to an entity type.

↳ Attribute

↳ Property of an entity type

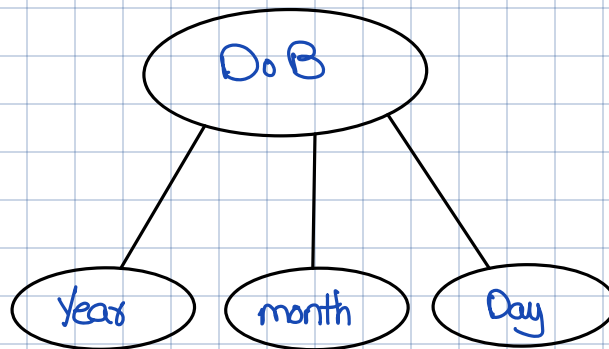
↳ Following are the types:

1. Simple



2. Composite

↳ Made up of simple attributes



3. Single Valued

↳ Holds only one value

4. Multi-valued

↳ Holds multiple values



5. Stored attributes

↳ Ones stored in the table

6. Derived

↳ Can be derived from the stored attributes
↳ If we have DoB, we can derive age.



7. Complex

↳ Combination of multi-value and composite attributes

8. Null Value Attribute:

↳ Missing value

↳ What should be an entity?

↳ When an object will have multiple instances

↳ When an entity & when an attribute?

↳ There is a fine line b/w.

↳ A tire manufacturer will call this an entity as it will need to maintain a lot of information on tires.

↳ A car manufacturer, on the other hand, will consider the tire as an attribute. It won't care how or where it was made.

↳ Keys

↳ It uniquely identifies an object

↳ There can be multiple of these

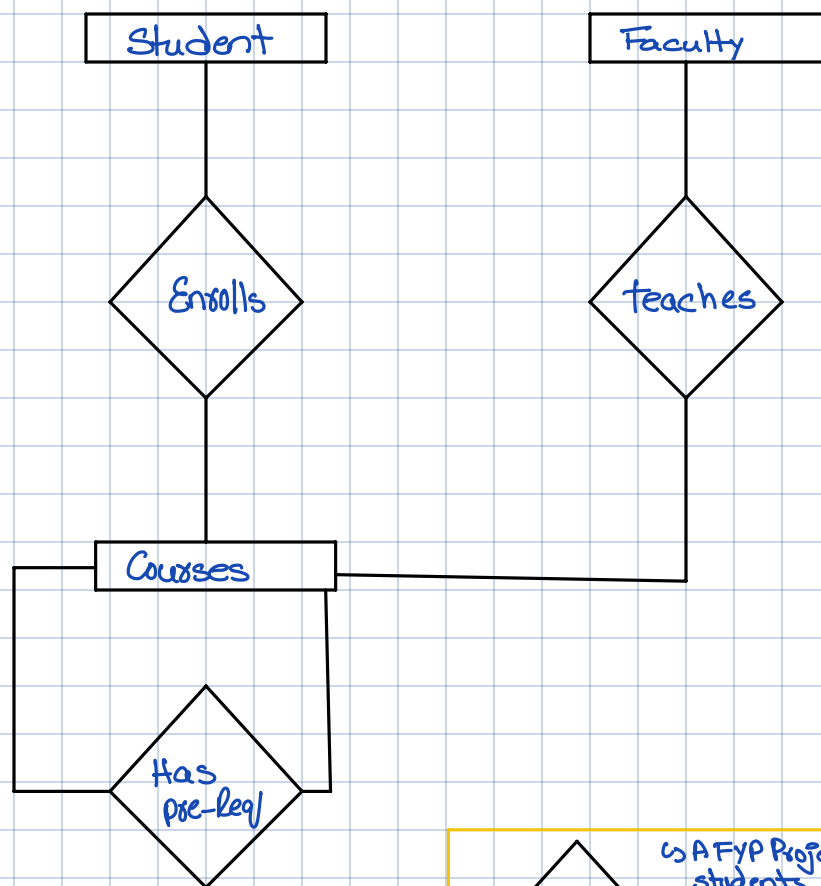
↳ When there are, it will be known as a composite key

StudentID

↳ Should not be NULL

Relationships:

- ↳ Defines relationship b/w two or more entity
- ↳ Entity is a noun and relationships are verbs



Degree of relationship:

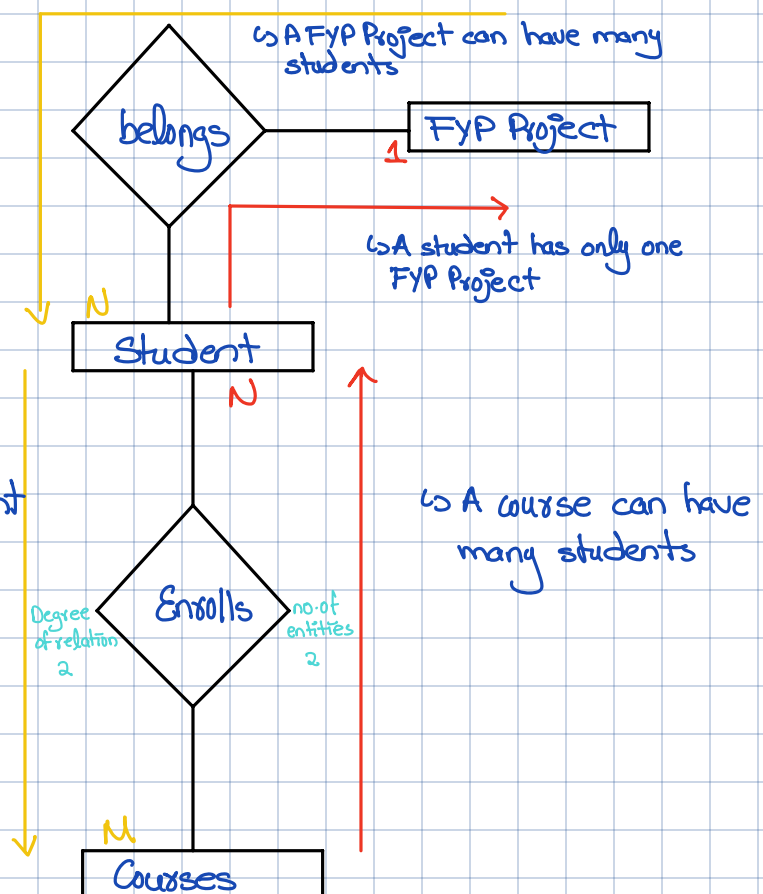
- ↳ There are two types:

1. Cardinality

- ↳ One to many
- ↳ Many to One
- ↳ Many to Many
- ↳ one to one

- ↳ Only for binary relationships

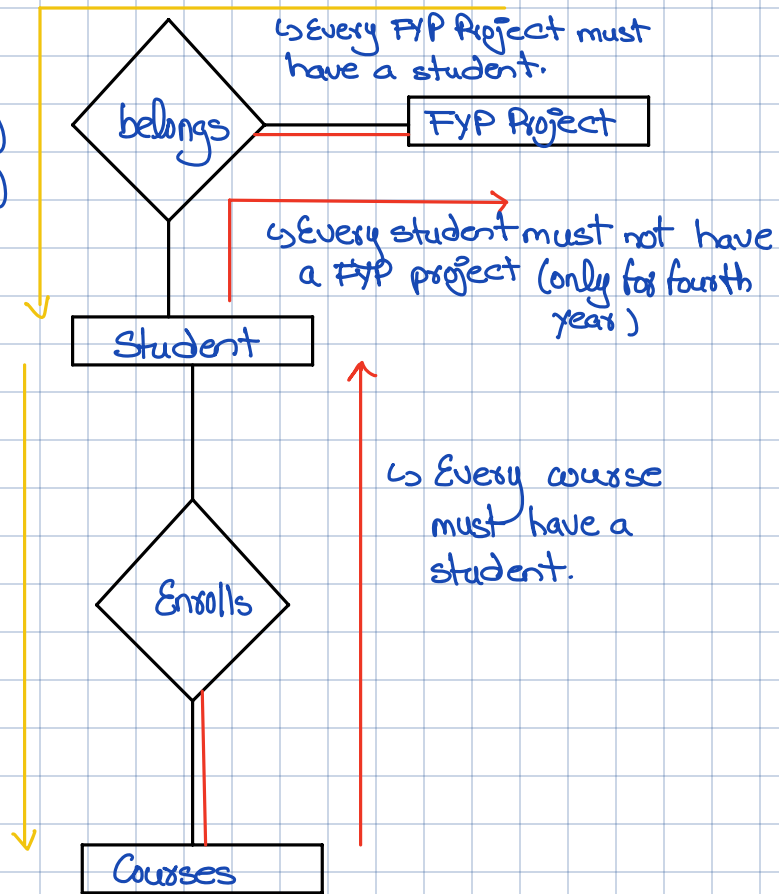
↳ A student can enroll in many courses



2. Participation:

↳ Either must participate (Total) completely or optional (Partial)

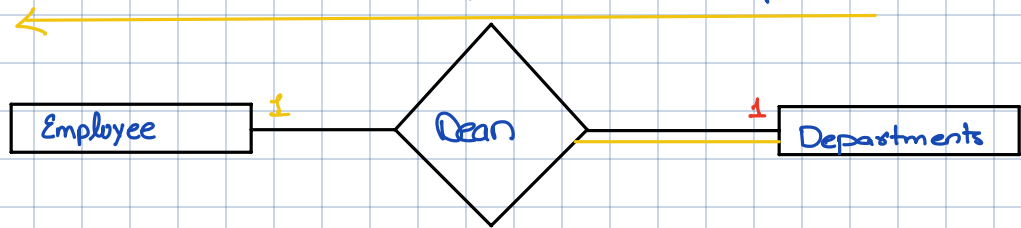
Every student must not be in a course.



↳ Weak Entity

↳ It's existence is dependent on another entity.

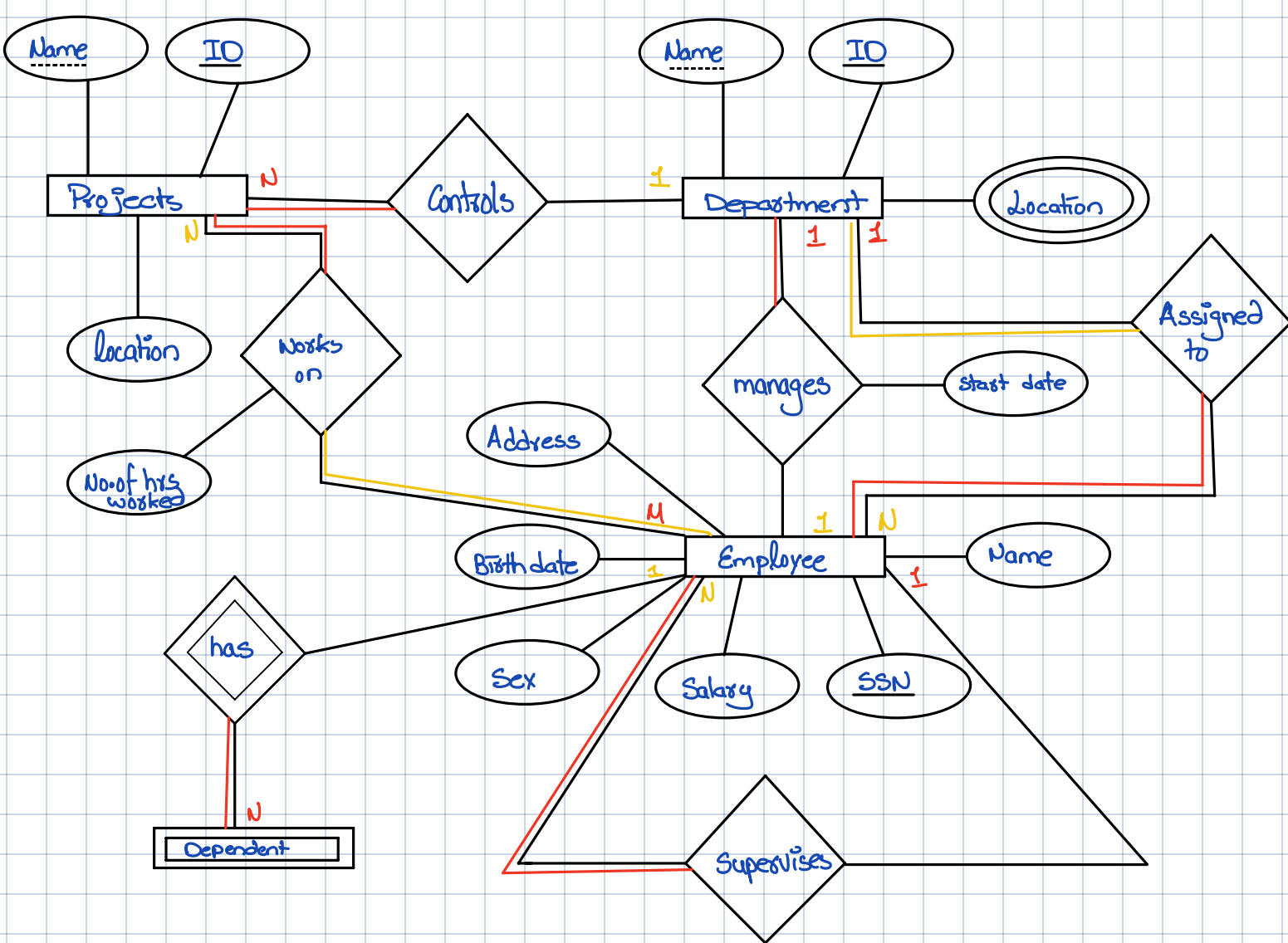
↳ Every Department must have a employee as dean?
↳ A Department has multiple deans



↳ A employee can be dean of many department

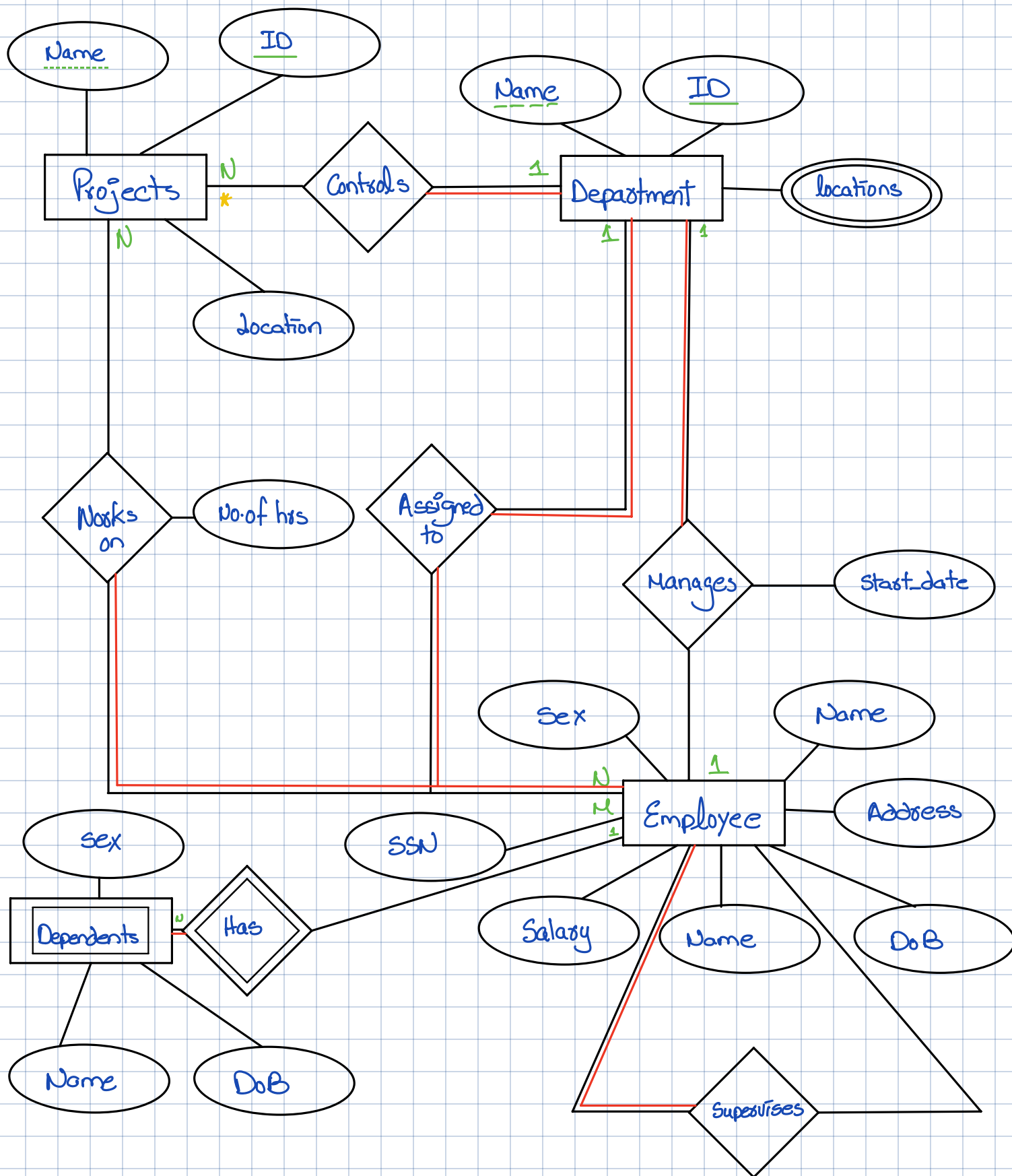
↳ Every employee must be a dean of a Department? No.

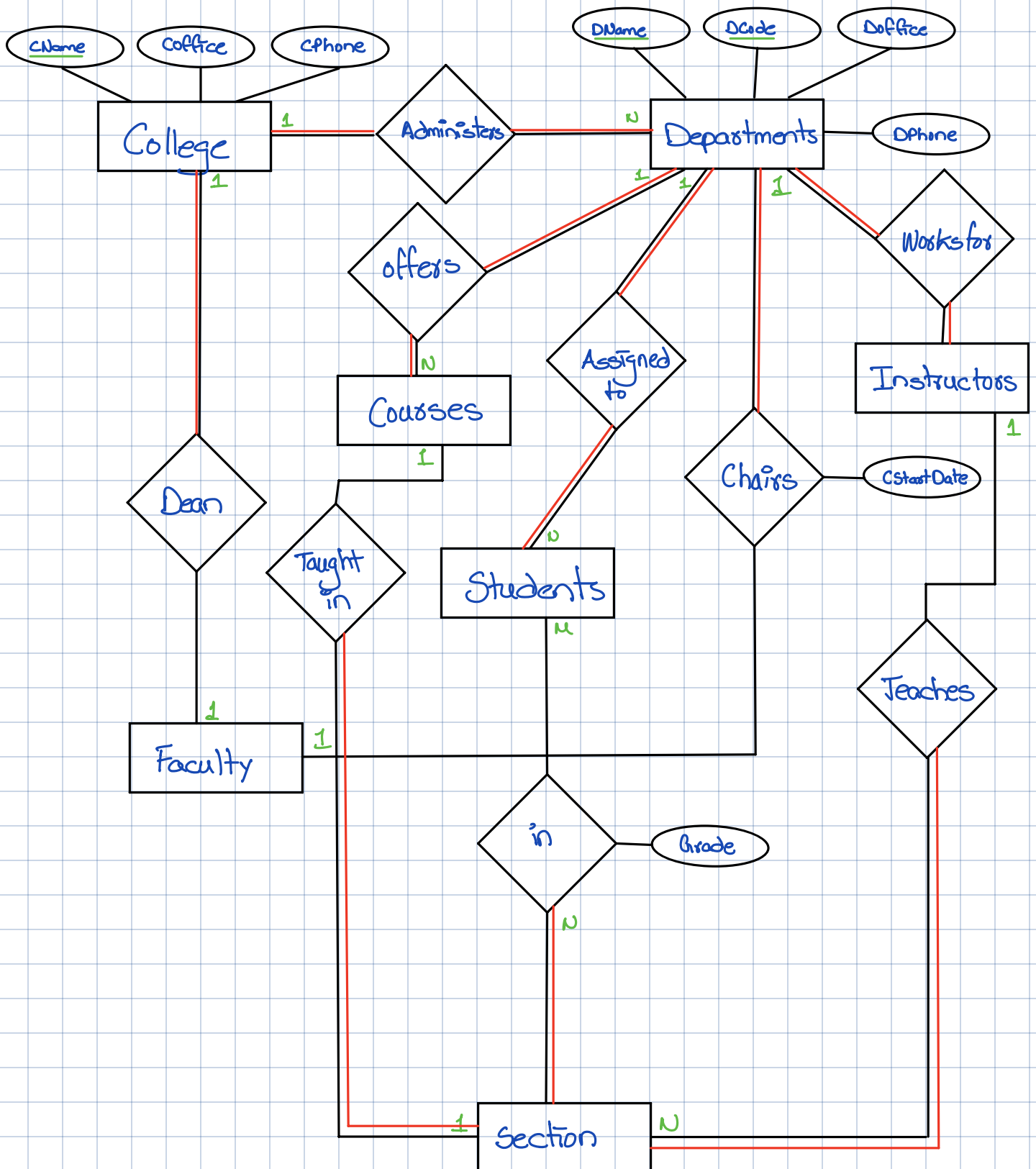
Case study 01:



PRACTICE:

Projects can be abandoned



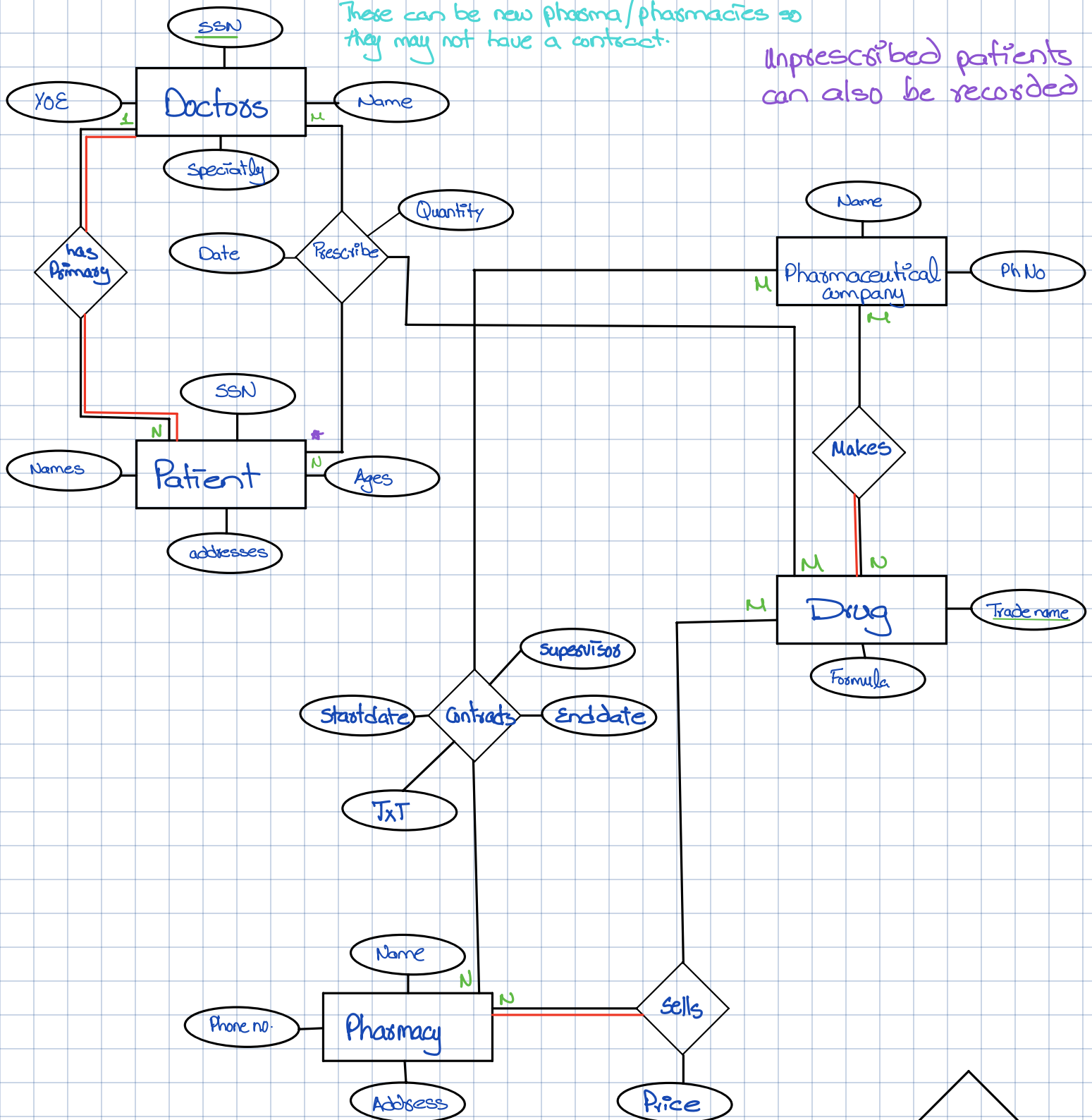


Assignment:

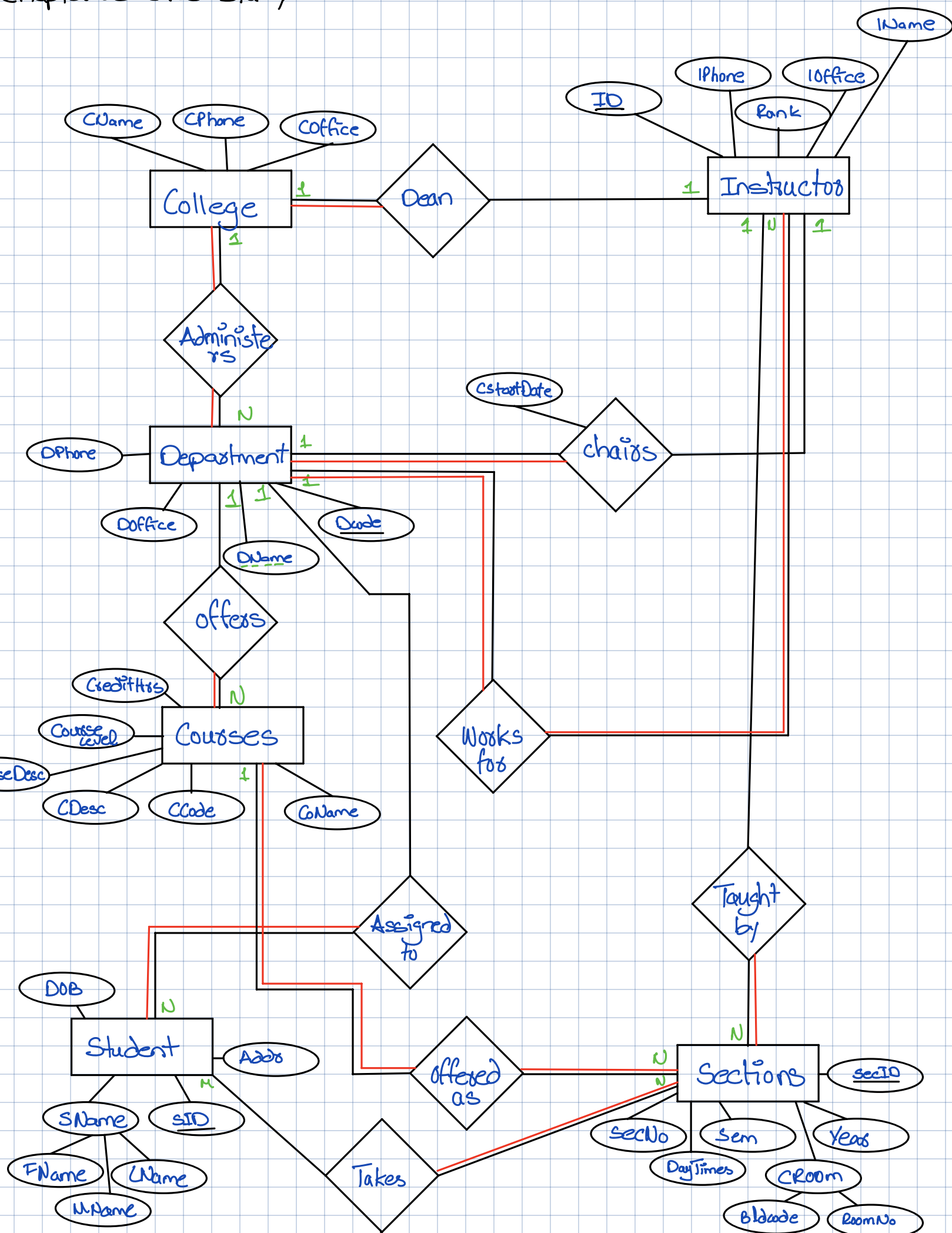
A Doctor in this scenario must participate fully

These can be new pharma/pharmacies so they may not have a contract.

Unprescribed patients can also be recorded



Chapter 03 case study



Takes